

Linux Web Server Log Analysis

ForensicLogX — BS Digital Forensics & Cyber Security

Case ID	FLX-20260603-232609	Date/Time	2026-06-03 23:26:09
Analyst	Gohar Ali	Classification	CONFIDENTIAL
Evidence File	test_log.txt	SHA-256 (first 16)	df63262a0449edc2...

1. Executive Summary

Analysis of **3** log entries from **test_log.txt** revealed **3** security threats, including **0** critical and **2** high severity events. The server processed requests from **2** unique IP addresses with an error rate of **66.67%**. Evidence of brute force attacks, potential DDoS flooding, directory traversal, and automated scanning activity was detected. Immediate remediation is recommended.

Metric	Value	Metric	Value
Total Requests	3	Threats Detected	3
Unique IPs	2	Critical Events	0
Error Rate	66.67%	Total Bytes	4,850

2. Threat Detection Findings

#	Type	Severity	Source IP	Count	Detail
1	Directory Traversal	MEDIUM	1.2.3.4	1	Path traversal attempt: /etc/passwd
2	Brute Force	HIGH	127.0.0.1	1	Detected from Uploaded Logs: Brute Force: Login attempts or
3	File Inclusion	HIGH	1.2.3.4	1	Detected from Uploaded Logs: LFI: Path Traversal

3. Chain of Custody

The following audit trail documents all actions performed on evidence file **test_log.txt** during this investigation.

#	Timestamp	Action	Detail	Actor
1	2026-06-03T23:26:09	Evidence Collected	File 'test_log.txt' collected for analysis	System
2	2026-06-03T23:26:09	Hash Computed	SHA-256: df63262a0449edc289d9f09beac43793b63beca8916a8ba09796c1274cc599cb	System
3	2026-06-03T23:26:09	Log Parsing Complete	3 entries extracted	Gohar Ali
4	2026-06-03T23:26:09	Threat Detection Complete	1 threats found	Gohar Ali
5	2026-06-03T23:26:09	Forensic Report Compilation	3 total threats identified	Gohar Ali

4. File Integrity Verification

Evidence File: test_log.txt

Algorithm: SHA-256

Hash Value:

df63262a0449edc289d9f09beac43793b63beca8916a8ba09796c1274cc599cb

5. Recommendations

- Immediately block identified malicious IPs using the generated iptables/ufw rules: 1.2.3.4, 127.0.0.1.
- Implement rate limiting on authentication endpoints (/wp-login.php, /admin, /login).
- Enable ModSecurity (Apache WAF) with OWASP Core Rule Set.
- Configure fail2ban to auto-block IPs after 5 consecutive authentication failures.
- Set up centralized log forwarding to a SIEM (e.g., ELK Stack, Wazuh) for real-time monitoring.
- Enforce HTTPS-only and disable directory listing on the web server.
- Schedule weekly automated log analysis runs with integrity verification.